Batch Least Squares Orbit Determination of a Low Earth Orbit Satellite from Range and Range Rate Tracking Data

MAE 182: Spacecraft Guidance and Navigation Term Project

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Abstract:

This report presents a batch least squares orbit determination program for a low Earth orbit satellite tracked by three stations using range and range rate measurements. The estimated state has 18 elements, the satellite position and velocity, the gravitational parameter μ, the second zonal harmonic J_2, the drag coefficient C_D, and the coordinates of two ground stations. The third station is held fixed to anchor the reference frame, since estimating all three station positions along with the satellite state makes the normal equations singular. The force model includes two body gravity, J_2 oblateness, and exponential atmospheric drag, and the state transition matrix is propagated alongside the trajectory at each iteration. Starting from the given approximate initial conditions, the batch converges in three iterations, dropping the range residual RMS from 733 m to 9.7 mm and the range rate RMS to 0.998 mm/s, both matching the noise levels of the simulated data. The post fit residuals are white with no remaining structure, the recovered parameters are physically reasonable, and the position error covariance gives a millimeter to centimeter 1σ error ellipsoid. The report also discusses why one station needs to be fixed, how the a priori covariance and data noise weighting affect the solution, and the numerical conditioning of the normal equations.

Introduction:

Orbit determination is the process of estimating the trajectory of a spacecraft from a set of imperfect tracking observations. Because the equations of motion are nonlinear and the measurements are corrupted by noise, the trajectory cannot be recovered exactly. Instead, a statistically optimal estimate is sought that best reconciles a dynamical model with the data. This report addresses the orbit determination of a near circular low Earth orbit (or LEO) patterned after NASA's QuikSCAT mission (altitude ≈ 800 km, inclination ≈ 98.6°) using the batch least squares algorithm. Three tracking stations provide range ρ and range rate $\dot{\rho}$ observations. One, a ship in the Pacific Ocean (station 101) and ground stations at Pirinclik, two, Turkey (337) and Thule, and three, Greenland (394). The data set spans approximately five hours and contains 385 observation epochs, with zero mean Gaussian noise of standard deviation 1 cm added to range and 1 mm s^{-1} added to range rate. The objective is to estimate the 18 element state

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{r}^{\boldsymbol{T}} & \dot{\boldsymbol{r}} & \mu & J_2 & C_D & \boldsymbol{x}_{\boldsymbol{s}_1}^{\boldsymbol{T}} & \boldsymbol{x}_{\boldsymbol{s}_2}^{\boldsymbol{T}} & \boldsymbol{x}_{\boldsymbol{s}_3}^{\boldsymbol{T}} \end{bmatrix}^T$$

where $\boldsymbol{r}$ and $\dot{\boldsymbol{r}}$ are the inertial (ECI) position and velocity, μ is the Earth gravitational parameter, J_2 is the second degree zonal gravity harmonic, C_D is the drag coefficient, and $\boldsymbol{x}_{\boldsymbol{s}_i}$ are the Earth fixed (ECEF) coordinates of station i. Because the coordinates of the moving ship (101) are treated as exact, station 101 is held fixed and only stations 337 and 394 are solved for. Table 1 lists the approximate initial values supplied for the problem.

Table 1: A priori (nominal) values supplied for the estimation

Quantity	***Value***	***Units***
$\boldsymbol{r_0}$	(757,700; 5,222,607; 4,851,500)	m
$\boldsymbol{\dot{r}_0}$	(2,213.21; 4,678.34; -5,371.30)	m/s
μ	$3.986004415 \times 10^{14}$	m^3/s^2
J_2	$1.082626925638815 \times 10^{-3}$	-
C_D	2.0	-
R_e	6,378,136.3	m
$\frac{A}{m}$	$\frac{3.0}{970}$	m^2/kg
$\dot{\theta}$	$7.2921158553 \times 10^{-5}$	rad/s
STATION 101	(-5,127,510; -3,794,160; 0)	m
STATION 337	(3,860,910; 3,238,490; 3,898,094)	m
STATION 394	(549,505; -1,380,872; 6,182,197)	m

Theory:

Dynamical Model

The satellite acceleration is modeled as the sum of two body gravity, the J2 oblateness perturbation, and atmospheric drag,

$$\ddot{\boldsymbol{r}} = \ddot{\boldsymbol{r}}_{\boldsymbol{2B}} + \ddot{\boldsymbol{r}}_{\boldsymbol{J_2}} + \ddot{\boldsymbol{r}}_{\boldsymbol{drag}} \tag{2}$$

The gravitational acceleration is the gradient of the potential including the J_2 term,

$$U = \frac{\mu}{r} - \frac{3\mu J_2 R_e^2}{2}\frac{z^2}{r^5} + \frac{\mu J_2 R_e^2}{2}\frac{1}{r^3}\ , \quad r = \|\boldsymbol{r}\|, \tag{3}$$

Whose gradient $\ddot{\boldsymbol{r}}_{\boldsymbol{gravity}} = \nabla$U, gives,

$$\ddot{x} = -\frac{\mu x}{r^3}\left[1 - \frac{3}{2}J_2\left(\frac{R_e}{r}\right)^2(5\frac{z^2}{r^2} - 1)\right]$$
$$\ddot{y} = -\frac{\mu y}{r^3}\left[1 - \frac{3}{2}J_2\left(\frac{R_e}{r}\right)^2(5\frac{z^2}{r^2} - 1)\right] \tag{4}$$
$$\ddot{z} = -\frac{\mu z}{r^3}\left[1 - \frac{3}{2}J_2\left(\frac{R_e}{r}\right)^2(5\frac{z^2}{r^2} - 3)\right]$$

The first term in each bracket is the central two body acceleration and the second is the J_2 correction. The sign of the J_2 term follows directly from the gradient of ***Equation (3)***; an early implementation error in this sign was identified and corrected during verification (see Results section).

Drag is modeled with an exponential density profile and a rotating atmosphere,

$$\ddot{\boldsymbol{r}}_{drag} = -\frac{1}{2}C_D\left(\frac{A}{m}\right)\rho_A v_A \boldsymbol{v_A}, \qquad \rho_A = \rho_0 \exp\left(-\frac{r-r_0}{H}\right), \tag{5}$$

Where the velocity relative to the corotating atmosphere is,

$$\boldsymbol{v_A} = \begin{bmatrix} \dot{x} + \dot{\theta}_y \\ \dot{y} - \dot{\theta}_x \\ \dot{z} \end{bmatrix}, \qquad v_A = \|\boldsymbol{v_A}\|, \tag{6}$$

with $\rho_0 = 3.614 \times 10^{-13}\ kg\ m^{-3}$, $r_0 = 700{,}000 + R_e$, $H = 88{,}667\ m$, and $\dot{\theta}$ the Earth rotation rate. At the ≈800 km altitude of this orbit the drag acceleration is many orders of magnitude smaller than the gravitational terms, but it is retained because C_D is an estimated parameter.

Lastly, in the state transition matrix, The estimator requires the sensitivity of the trajectory to its initial state. The dynamic subset $\boldsymbol{x_d} = [\boldsymbol{r}; \dot{\boldsymbol{r}}; \mu; J_2; C_D]$ evolves as $\dot{\boldsymbol{x}}_d = \boldsymbol{f}(\boldsymbol{x_d})$, and the 9x9 state transition matrix $\Phi(t, t_0)$ satisfies the variational equation,

$$\dot{\Phi}(t,t_0) = A(t)\Phi(t,t_0), \qquad A(t) = \frac{\partial f}{\partial x_d}(\boldsymbol{X}^*), \qquad \Phi(t_0,t_0) = I_9 \tag{7}$$

The Jacobian A contains the partial derivatives of the acceleration with respect to position, velocity, and the parameters (μ, J_2, C_D). The station coordinates are constants of the motion and do not appear in the propagated STM (their block of the full 18x18 STM is the identity). The trajectory and the 81 STM elements are integrated simultaneously with a variable step Runge Kutta scheme (ode45) at relative and absolute tolerances of 10^{-12}.

Observation Model

Each station measures the range and range rate to the satellite. With the satellite expressed in ECI and the station in ECEF, the Earth rotation angle is $\theta = \dot{\theta}t$ and the predicted range is

$$\rho = [x^2 + y^2 + z^2 + x_s^2 + y_s^2 + z_s^2 - 2(xx_s + yy_s)cos\theta + 2(xy_s - yx_s)sin\theta - 2zz_s]^{1/2}, \tag{8}$$

And the predicted range rate is

$$\dot{\rho} = \frac{1}{\rho}[x\dot{x} + y\dot{y} + z\dot{z} - (\dot{x}x_s + \dot{y}y_s)cos\theta + \dot{\theta}(xx_s + yy_s)sin\theta + (\dot{x}y_s - \dot{y}x_s)sin\theta + \dot{\theta}(xy_s - yx_s)\cos\theta - z\dot{z}_s], \tag{9}$$

where (x_s, y_s, z_s) are the ECEF coordinates of the observing station. The observation state mapping matrix $\widetilde{H}$ collects the partial derivatives of the two observables with respect to the full state,

$$\widetilde{H}(t) = \begin{bmatrix} \frac{\partial \rho}{\partial \boldsymbol{X}} \\ \frac{\partial \dot{\rho}}{\partial \boldsymbol{X}} \end{bmatrix} \in \mathbb{R}^{2\times 18} \tag{10}$$

The observables depend on the satellite position and velocity and on the observing station's coordinates, but not directly on μ, J_2, or C_D; the corresponding columns of $\widetilde{H}$ are therefore zero, and the sensitivity of the measurements to those parameters enters only through the STM.

Batch Least Squares Algorithm

The batch processor estimates a single state correction at the epoch t_0 that best fits the entire observation arc in a weighted least squares sense. Linearizing the observation $\boldsymbol{Y_i}$ about the reference trajectory gives the residual $\boldsymbol{y_i} = \boldsymbol{Y_i} - \boldsymbol{G}(\boldsymbol{X_i^*}, t_i)$ and the relation $\boldsymbol{y_i} \approx \widetilde{H_i}\delta\boldsymbol{X_i} + \boldsymbol{\varepsilon_i}$. Mapping the state deviation back to the epoch with the STM, $\delta\boldsymbol{X_i} = \Phi(t_i, t_0)\delta\boldsymbol{x_0}$, yields the epoch referenced observation matrix

$$H_i = \widetilde{H}_i\Phi(t_i, t_0) \tag{11}$$

The maximum likelihood (and minimum variance) estimate of the epoch deviation $\widehat{\boldsymbol{x}}_0$ that incorporates an a priori estimate $\overline{\boldsymbol{x}}_{\boldsymbol{0}}$ with covariance $\bar{P}_0$ solves the normal equations

$$\left(\bar{P}_0^{-1} + \sum_{i=1}^{\ell} H_i^T R^{-1} H_i\right)\widehat{\boldsymbol{x}}_{\boldsymbol{0}} = \bar{P}_0^{-1}\overline{\boldsymbol{x}}_{\boldsymbol{0}} + \sum_{i=1}^{\ell} H_i^T R^{-1}\boldsymbol{y_i}$$

Where $R = diag(\sigma_\rho^2, \sigma_{\dot{\rho}}^2)$ is the measurement noise covariance. The solution and its covariance are

$$\widehat{\boldsymbol{x}}_0 = \Lambda^{-1}\boldsymbol{N}, \qquad P_0 = \Lambda^{-1} \tag{13}$$

Because the problem is nonlinear, the estimate is iterated: the nominal state is updated, $\boldsymbol{X_0^*} \leftarrow \boldsymbol{X_0^*} + \widehat{\boldsymbol{x}}_{\boldsymbol{0}}$, the a priori deviation is shifted, $\overline{\boldsymbol{x}}_{\boldsymbol{0}} \leftarrow \overline{\boldsymbol{x}}_0 - \widehat{\boldsymbol{x}}_0$, and the trajectory, STM, and residuals are recomputed until the residual RMS stops decreasing.

A priori and noise statistics. The a priori covariance follows the project specification,

$$\bar{P}_0 = diag(10^6\, I_3, 10^6\, I_3, 10^{20}, 10^6, 10^6, 10^{-10}\, I_3, 10^6 I_3, 10^6 I_3), \tag{14}$$

in SI units, ordered as in Eq. (1). The very small variance (10^{-10}) on station 101 coordinates pins that station, while the very large variance (10^{20}) on μ leaves it effectively unconstrained by the prior. The measurement weighting is $R = diag(\sigma_\rho^2, \sigma_{\dot{\rho}}^2)$ with $\sigma_\rho = 1\, cm$ and $\sigma_{\dot{\rho}} = 1\, mm\, s^{-1}$.

Software:

The program is written in MATLAB and split into a few focused files. A single setup function (*project_setup.m*) holds all the constants, initial conditions, a priori statistics, and observation data, so every part of the estimator is working from the same numbers. The dynamics and STM derivatives live in *project_dyn.m*, which calls *computed_A.m* for the Jacobian and the observation model and its partials are handled in *computed_Htilde.m*. The driver script (*main.m*) runs the batch filter loop and generates all the results and figures. Each iteration integrates the trajectory and STM over the full arc, then samples it every 20 seconds to line up with the observation times. The whole thing runs from one script with no external dependencies.

Results:

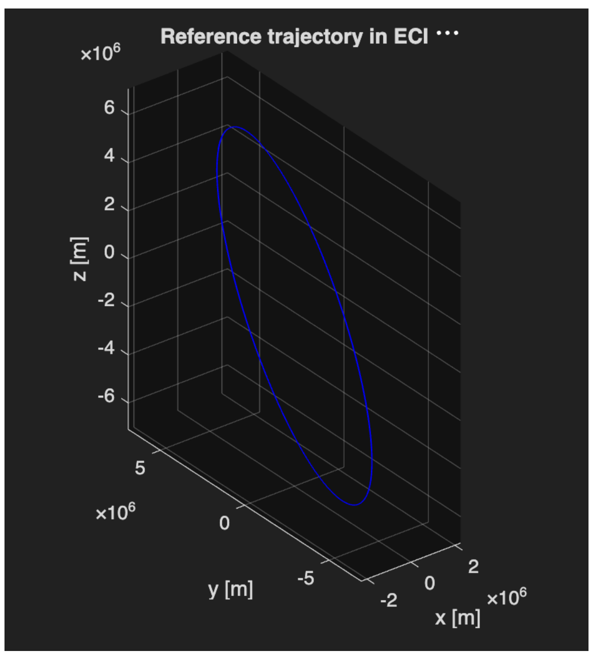

Figure 1: Reference satellite trajectory in the ECI frame ($\approx 800\ km$, 98.6°inclination), produced by integrating the two-body + J2 + drag model.

The reference trajectory, shown in Figure 1, is the expected near polar LEO ellipse. This confirms the equations of motion are integrating correctly. The propagated state and linearization matrices were checked against the Part I requirements and the final position and velocity came out to $\boldsymbol{r}(t_f) = (1.1292, 5.9911, 3.7735) \times 10^6\ m$ and $\dot{\boldsymbol{r}}(t_f) = 2.0088, 3.5610, -6.2388) \times 10^3 m/s$

During verification, the first iteration range residual RMS was two orders of magnitude larger than expected. The problem turned out to be a sign error on the J2 acceleration term in the dynamics, which was inconsistent with the Jacobian in computed_A.m. Fixing the sign to match the gradient brought the first iteration range RMS down to 733 m, consistent with a correct linearization, and the estimator converged from there.

Table 2: Pre fit through converged residual RMS by iteration

Iteration	**Range RMS (m)**	**Range Rate RMS (m/s)**
1	7.3275×10^2	2.9002
2	3.1957×10^{-1}	1.1997×10^{-3}
3	9.7249×10^{-3}	9.9792×10^{-4}
4	9.7249×10^{-3}	9.9792×10^{-4}
5	9.7249×10^{-3}	9.9792×10^{-4}

Table 2 above lists the weighted residual RMS at each iteration. Starting from the supplied nominal, the range RMS falls from 733 m to 9.7 mm and the range rate RMS from 2.9 $m\ s^{-1}$ to 0.998 $mm\ s^{-1}$ within three iterations, after which both are unchanged. The converged values equal the measurement noise standard deviations (1 cm and 1 $mm\ s^{-1}$), which is the best achievable fit. The estimator has extracted all available information and the residuals are reduced to noise. Figures 2a and 2b show the RMS history on a logarithmic scale.

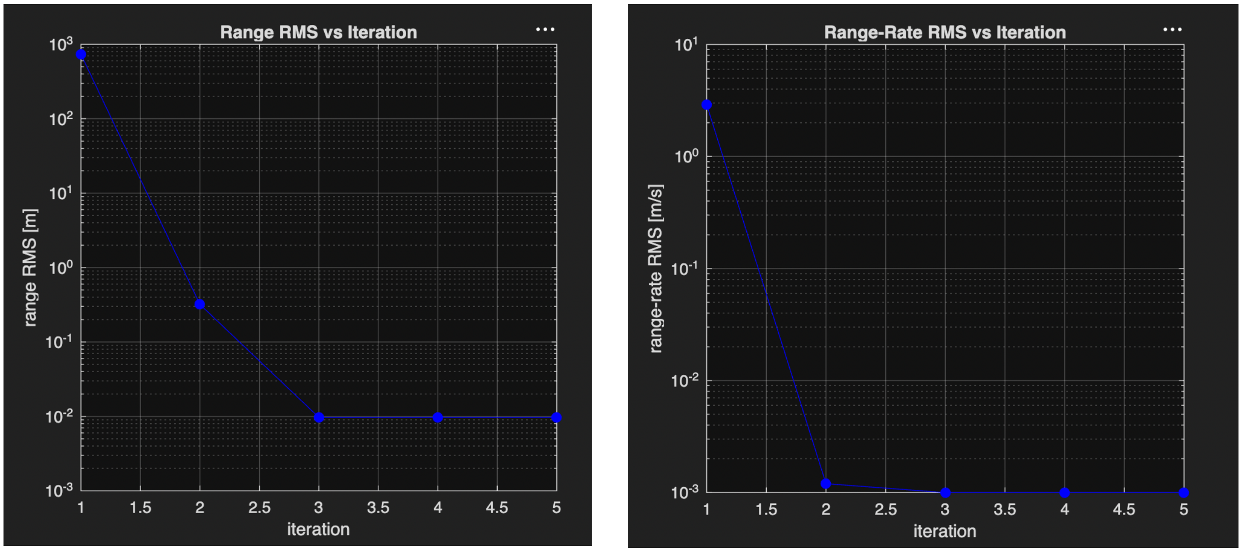

Figure 2a and Figure 2b: Weighted residual RMS versus iteration (log scale). Both data types converge to the measurement noise floor by the third iteration.

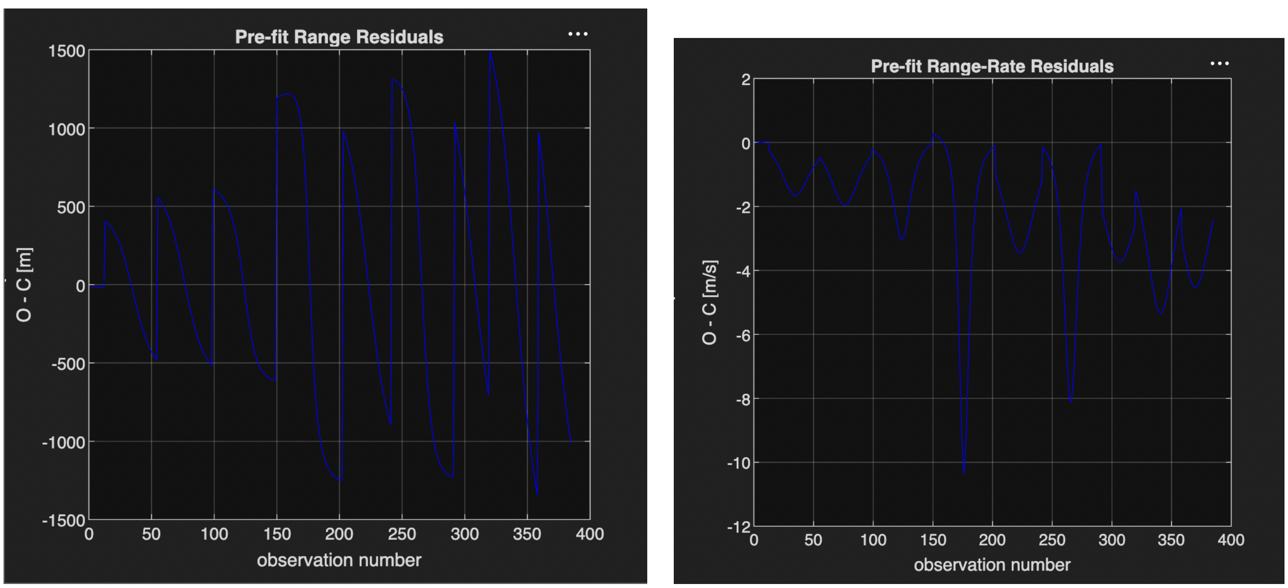

Figure 3: Pre-fit residuals about the initial nominal trajectory, showing systematic per pass structure of order 1000 m (range) driven by the initial state and parameter errors.

Figure 3 shows the pre fit residuals, computed against the initial nominal trajectory. They have large, systematic swings on the order of 1000 m in range that track the observation passes. This is a clear sign that the reference state and dynamics have an error. After convergence, the post fit residuals (Figure 4) are reduced to zero mean noise within $\pm 0.02\ m$ in range and $\pm 0.002\ m\ s^{-1}$ in range rate. The transition from structured pre fit residuals to white post fit residuals is the main qualitative sign that the estimator has converged correctly and that no unmodeled signal remains.

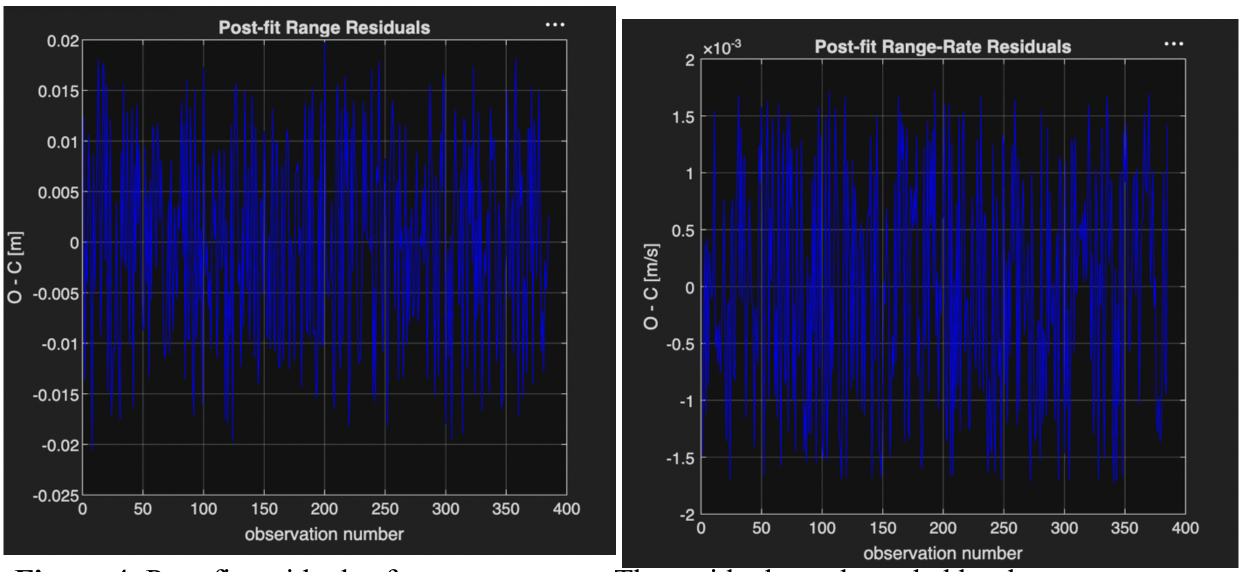

Figure 4: Post-fit residuals after convergence. The residuals are bounded by the measurement noise level ($\pm 0.02\ m, \pm 0.002\ m\ s^{-1}$), indicating that all observable signal has been fit.

Table 3 reports the converged estimate and the 1σ uncertainties from P_0. The satellite position and velocity move by less than 0.5 m and 0.05 $m\ s^{-1}$ from the supplied nominal. This confirms that the provided initial conditions were already accurate. The estimated J_2 and μ remain within their priors, and C_D settles at 2.19. The solved station coordinates for 337 and 394 move by several meters from their nominal positions. This reflects the information the data provides about their locations. The recovered 1σ position uncertainties are at the centimeter level.

Table 3: Converged batch estimate at epoch t_0 and 1σ uncertainties

Quantity	Estimate	1σ	Units
x	757,700.29	7.53×10^{-3}	m
y	5,222,606.578	1.18×10^{-2}	m
z	4,851,499.738	1.48×10^{-2}	m
$\dot{x}$	2,213.250618	8.64×10^{-6}	m/s
$\dot{y}$	4,678.372709	1.44×10^{-5}	m/s
$\dot{z}$	-5,371.314415	1.02×10^{-5}	m/s
μ	$3.98600399 \times 10^{14}$	4.16×10^{5}	m^3/s^2
J_2	$1.08199945 \times 10^{-3}$	2.45×10^{-10}	$-$
C_D	2.188709	3.81×10^{-3}	$-$
Station 337	(3,860,899.99, 3,238,500.00, 3,898,099.98)	$(5.2712, 8.4479, 8.7610) \times 10^{-3}$	m
Station 394	(549499.99, −1380869.98, 6182199.98)	$(7.3389 \times 10^{-3}, 1.2748 \times 10^{-2}, 1.6558 \times 10^{-2})$	m

Position Error Ellipsoid

The $3x3$ position block of the state covariance, $P_{rr} = P_0(1{:}3,1{:}3)$, defines the position error ellipsoid. Its principal axes are the eigenvectors of P_{rr} and the 1σ semi axis lengths are the square roots of the eigenvalues,

$$P_{rr} = VDV^T, \qquad a_k = \sqrt{D_{kk}} \tag{15}$$

The computed semi axes are (3.83, 6.66, 18.9)$\times 10^{-3}$, with a condition number $k(P_{rr}) = 24.3$. As Figure 5 shows ellipsoid is elongated along its major axis which indicates that position is less well observed in that particular direction given the tracking geometry, but remains subcentimeter to centimeter scale in all directions and well conditioned overall.

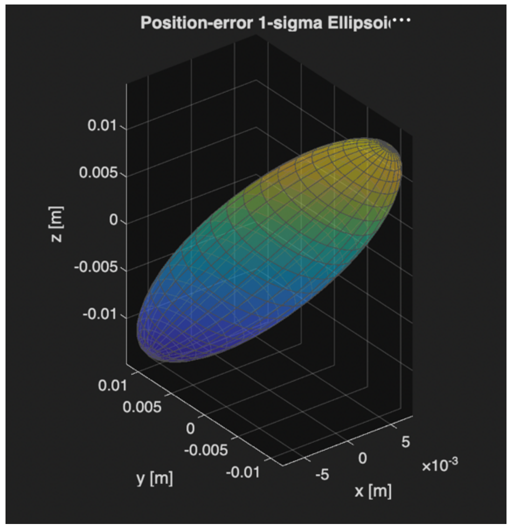

Figure 5: Position error 1σ ellipsoid from the batch covariance P_{rr}. Semi axes (3.83, 6.66, 18.9) mm; the elongation marks the least observable position direction.

Discussion

The range and range rate depend only on the relative geometry between the satellite and the observing station. In Equations (8) and (9) the satellite and station coordinates enter together. If the satellite state and all three station positions were estimated simultaneously from range data alone, then a problem would occur. The entire constellation of satellite orbit and stations could be translated and rotated as a rigid set without changing any measurement. The estimator would have no way to resolve the ambiguity, and the information matrix Λ would be singular. Holding one station fixed removes the ambiguity by anchoring the origin and orientation of the Earth fixed frame which renders the remaining state observable.

Station 101 is held fixed by giving it a near zero a priori variance (10^{-10}in Equation 14), which forces its estimate to stay at the nominal value. When that prior variance goes to zero, this is the same as dropping the station from the state vector and treating its coordinates as known constants. Both approaches give the same satellite and parameter estimates. The difference is practical. Removing the station shrinks the estimated state from 18 to 15 elements and gives you a smaller and better conditioned normal matrix. The near zero variance approach keeps the full 18 element state but introduces a very large entry (10^{10}) into Λ^{-1}. The fixed covariance form is used to keep a consistent 18 state interface across all estimators.

The fixed station defines the reference frame, so the choice is not arbitrary. Station 101 is the natural anchor because its coordinates are specified as exact, whereas the ground stations 337 and 394 are the quantities of interest to be refined. Equally important, the ship is in motion and its dynamics are deliberately not modeled. Fixing a moving platform's instantaneous coordinates while solving for the ground stations is the right arrangement. Fixing a different station would still anchor the frame, but the resulting solution and its covariance would differ because the three stations observe the orbit over different geometries and for different fractions of the arc. This gives them unequal abilities to constrain the trajectory.

The a priori covariance $\bar{P}_0$ sets the relative weight of the prior versus the data. A tight prior which means it has small variances, holds the estimate near the nominal and suppresses the influence of the observations. A loose prior lets the data dominate. For this data set the position, velocity, and ground station coordinates are observable, and the accumulated data information $\Sigma H_i^T R^{-1} H_i$ overwhelms the prior term$\bar{P}_0^{-1}$. This means the converged estimate of these states is essentially insensitive to $\bar{P}_0$ within reasonable bounds. The observable parameters μ and J_2 are the exception. Their large prior variances let the data move them, but they remain close to the reference because the five hour arc constrains them only weakly, and their reported uncertainties are dominated by the data geometry.

The measurement noise covariance R governs the relative weighting of range and range rate observations through $W = R^{-1} = diag\left(\sigma_{\rho}^{-2}, \sigma_{\dot{\rho}}^{-2}\right)$. Scaling R by a common factor does not change the estimate, it rescales Λ and **N** and only rescales the reported covariance. Changing the ratio $\frac{\sigma_{\rho}}{\sigma_{\dot{\rho}}}$ redistributes influence between the two data types. With the specified 1 cm and 1 $mm\ s^{-1}$ noise levels, both observation types contribute meaningfully and the post fit residuals of each settle at their respective noise levels which confirms a consistent weighting.

The specified prior is loose, the $10^6 m^2$ position variance corresponds to a 1σ uncertainty of 1000 m. This is larger than the true initial error, which the converged solution shows to be of order a meter. Using an a priori which is more compatible with the actual errors, like for example a few meters in position, would tighten the prior toward the truth and it would; first reduce the size of the first correction, second accelerate convergence, and finally, the most important, improve the numerical conditioning of the normal equations. The information matrix in this problem is ill conditioned because the estimated quantities span a large dynamic range, μ $\sim 10^{14}$ against parameters and coordinates of order unity to 10^6, because the near zero prior on station 101 has a 10^{10} term. The solution reaches the measurement noise floor, confirming that the conditioning degrades accuracy negligibly. A prior scaled to the true uncertainties, or a nondimensionalization of the state, would reduce the condition number.

Conclusion

A batch least squares orbit determination program was developed and verified for a QuikSCAT like low Earth orbit satellite tracked in range and range rate by three stations. The estimated state includes position, velocity, μ, J_2, C_D, and the coordinates of two tracking stations, for a total of 18 elements. The force model combines two body gravity, J_2 oblateness, and exponential atmospheric drag, and the state transition matrix is propagated alongside the trajectory at each iteration. Station 101 was held fixed to anchor the Earth fixed reference frame, since estimating all three station positions at the same time as the satellite state leads to a singular information matrix. Starting from the given approximate initial conditions, the batch converged in three iterations. The range residual RMS dropped from 733 m down to 9.7 mm and the range rate RMS reached 0.998 mm/s, both at the level of the noise in the data. The post fit residuals are white and free of structure, confirming the estimator worked correctly and no significant signal was left unmodeled. The recovered drag coefficient ($C_D \approx 2.19$) and the position error ellipsoid (semi axes 3.83, 6.66, 18.9 mm) are physically reasonable results. One of the most important lessons from building the program was that a sign error in the J_2 acceleration that did not match the Jacobian prevented convergence entirely, and fixing that inconsistency was the step that finally made the batch work. The analysis also showed that once the data information is large enough, the solution is not very sensitive to the choice of a priori covariance.